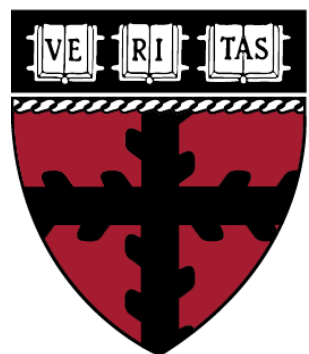


Coactive Learning for Large Language Models using Implicit User Feedback

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joint work with: Thorsten Joachims, Aaron Tucker, Adam Cahall



Harvard John A. Paulson
School of Engineering
and Applied Sciences



Kempner
INSTITUTE
at Harvard University

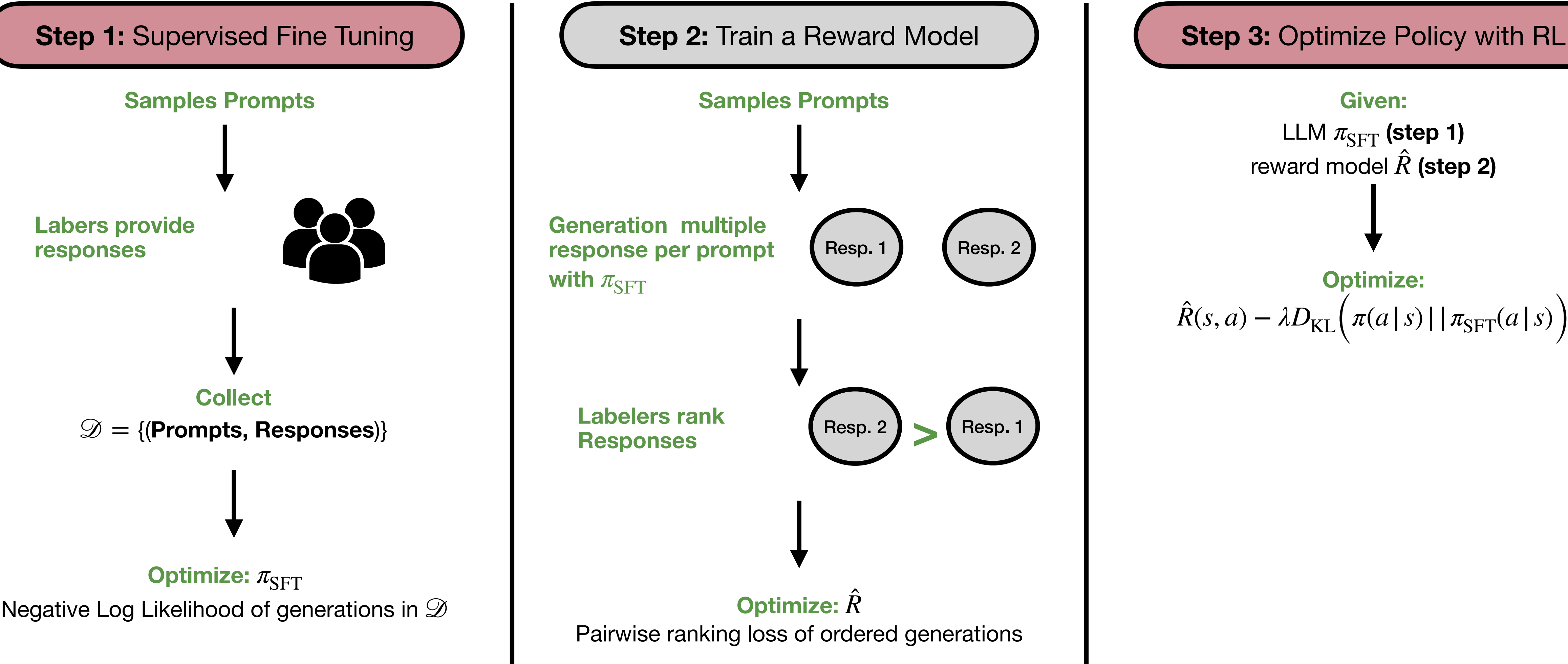
"A large language model is a deep learning model trained on massive text datasets to understand and generate human language."



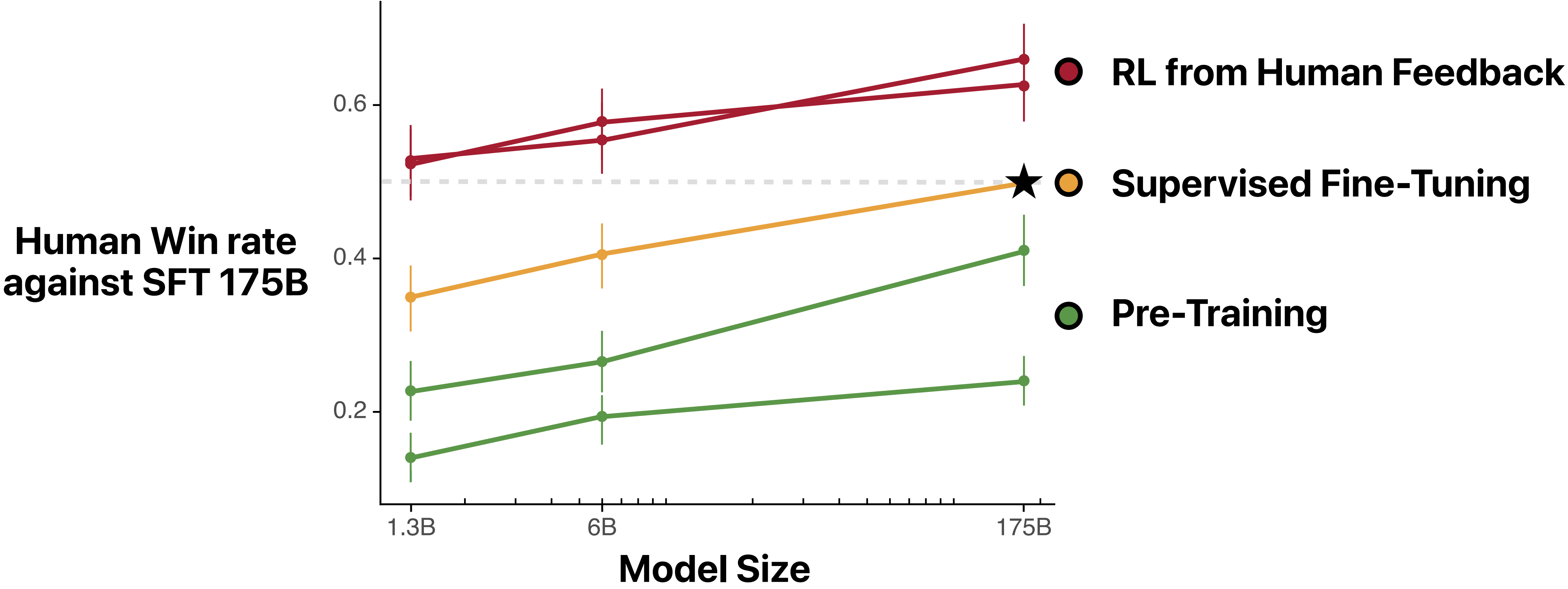
Large Language Model

How can we efficiently align LLMs to human preferences?

RL from Human Feedback (RLHF)



Learning from Human Feedback



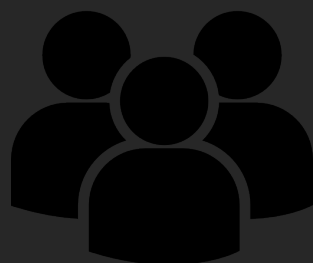
RL from Human Feedback (RLHF)

Step 1: Supervised Fine Tuning

Samples Prompts



Labors provide
one response



Collect

$\mathcal{D} = \{(\text{Prompts}, \text{Desired Generation})\}$



Optimize: π_{SFT}

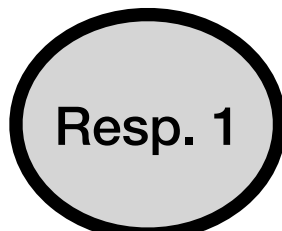
Negative Log Likelihood of generations in $\mathcal{D}$

Step 2: Train a Reward Model

Samples Prompts



Generation multiple
response per prompt
with π_{SFT}



Research Question:

Can implicit user edits be used as a reliable and efficient
signal for preference-based fine-tuning of LLMs?



Optimize:

Pairwise ranking loss of ordered generations

Step 3: Optimize Policy with RL

Given:

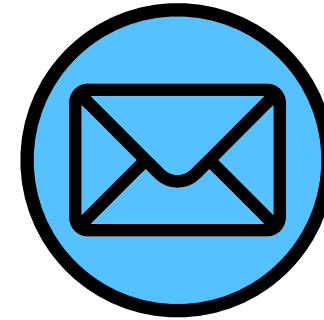
LLM from SFT (step 1)
reward $\hat{r}$ (step 2)



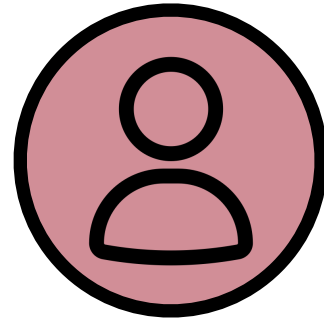
Optimize:

$$\hat{r}(s, a) - \lambda D_{\text{KL}}(\pi(a|s) || \pi^{\text{SFT}}(a|s))$$

Email Writing Assistant



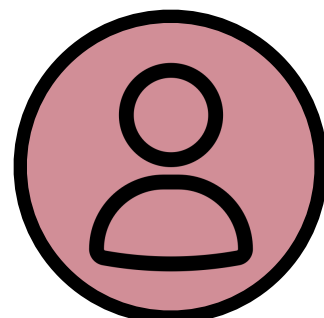
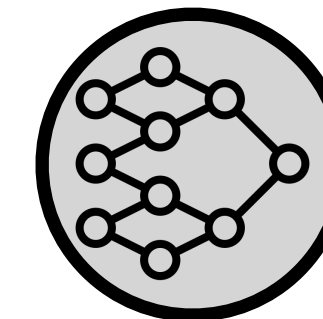
Just want to follow up on our lunch plan - I am available tomorrow, if this week's Wed works for you, or next week Mon and Tue; Otherwise, I'll be back on 11/29.



This week is a little rough, next Monday and tuesday are not great for me. Mondays I have lab meetings and research meeting around lunch time. Tuesday I have am teaching a course that is roughly around lunch time. Maybe lets reconnect wehn you are back 11/29.

Improve the text

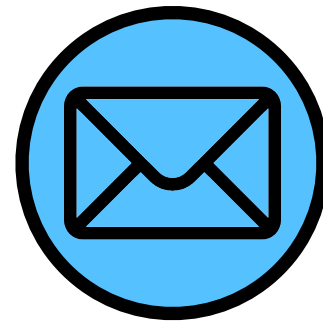
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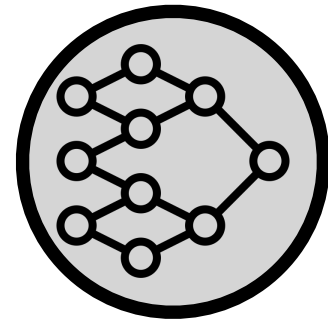
x_t



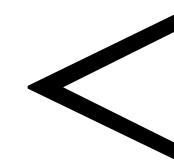
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y_t

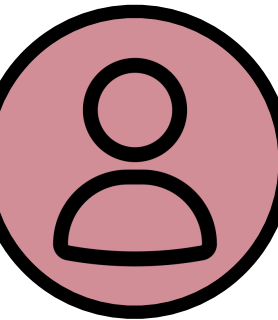
$\bar{y}_t$



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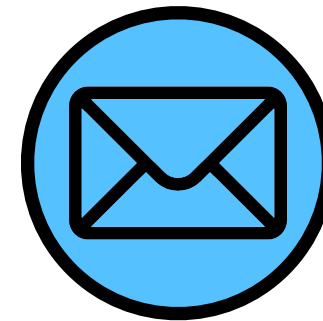


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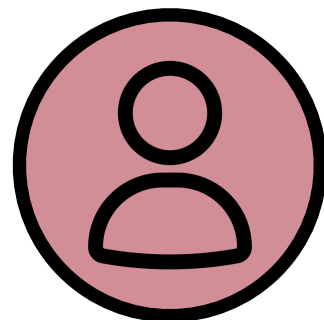
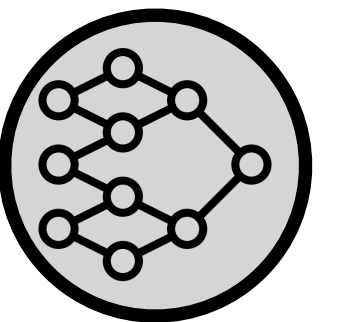
conventional rlhf



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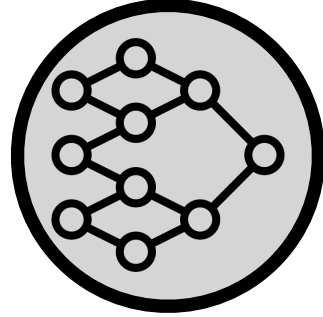
x_t



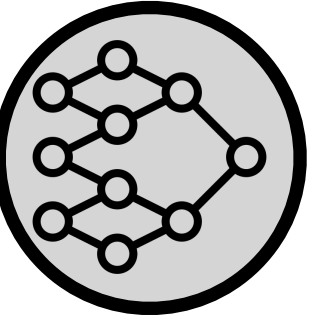
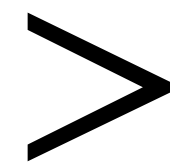
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y_t

$\bar{y}_t$

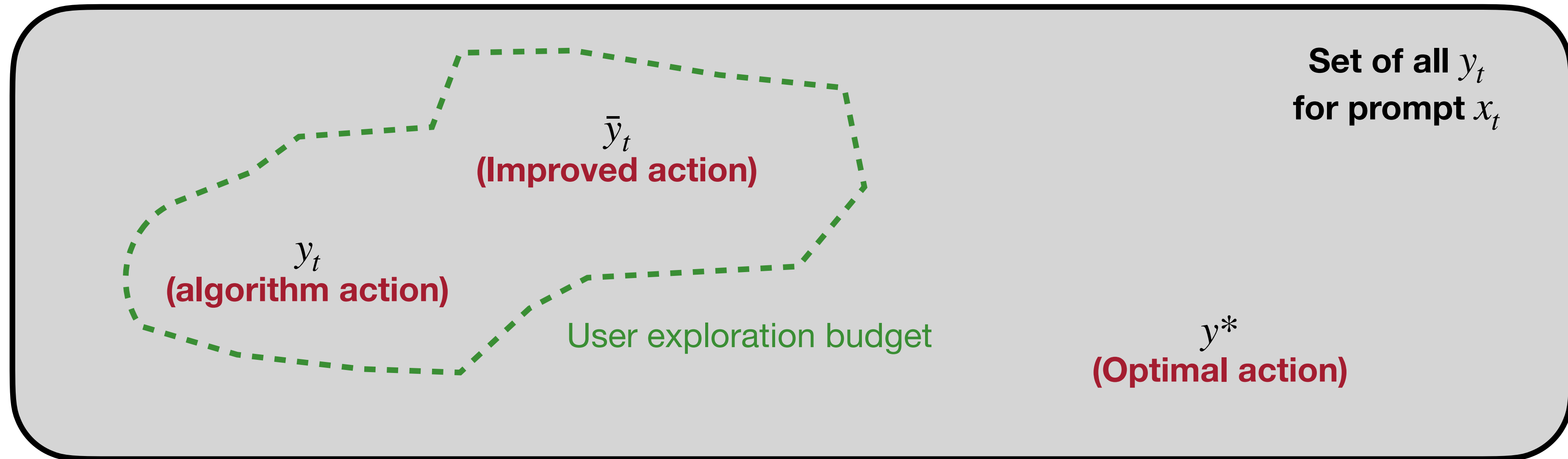
A circular icon containing a black and white diagram of a neural network with interconnected nodes.

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Coactive Feedback Model



Given prompt x_t

Algorithm action y_t

Improved action $\bar{y}_t : R(x_t, \bar{y}_t) > R(x_t, y_t)$

Optimal action $y_t^* = \arg \max_y R(x_t, y)$

Coactive Learning Model

Loop Forever

Observe context x (e.g., prompt)

Learning algorithm **presents** y (e.g., response)

User returns $\bar{y}$ with $R(x, \bar{y}) > R(x, y)$

Regret = $\text{Regret} + [R(x, y^*) - R(x, y)]$

Other Online Learning Models

Bandit setting: for selected y , receive $R(x, y)$

Dueling Bandit setting: for selected y **and** $\bar{y}$, receive $R(x, \bar{y}) > R(x, y)$

Coactive setting: for selected y , **receive** $\bar{y}$ and $R(x, \bar{y}) > R(x, y)$

Coactive Learning Regret Bound

Quantifying Feedback Quality

$$\underbrace{R^*(x_t, \bar{y}_t) - R^*(x_t, y_t)}_{\text{User feedback}} \geq \alpha \underbrace{\left(R^*(x_t, y_t^*) - R^*(x_t, y_t) \right)}_{\text{Optimal gap}} - \underbrace{\xi_t}_{\text{Model error}}$$

User feedback

Optimal gap

Model error

For α -informative feedback and pairwise learner A_{pair} with loss:

$$\bar{\Delta}(T \mid A_{\text{pair}}) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[R^*(y_t \mid x_t) > R^*(\bar{y}_t \mid x_t)] [R^*(\bar{y}_t \mid x_t) - R^*(y_t \mid x_t)]$$

Regret bound:

$$\frac{1}{T} \sum_{t=1}^T [R^*(y_t^* \mid x_t) - R^*(y_t \mid x_t)] \leq \underbrace{\frac{\bar{\Delta}(T \mid A_{\text{pair}})}{\alpha T}}_{\rightarrow \text{zero}} + \underbrace{\frac{1}{\alpha T} \sum_{t=1}^T \xi_t}_{\text{Model error}}$$

Email Writing Assistant

conventional rlhf

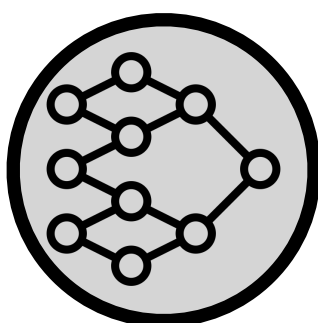
x_t



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y_t

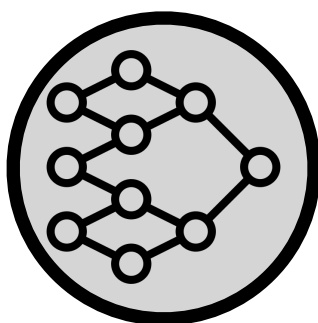
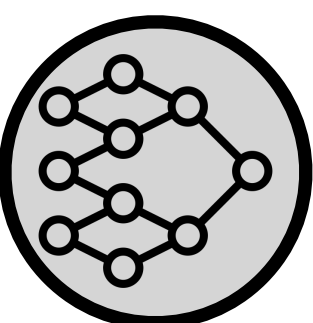
$\bar{y}_t$



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Explicit Feedback

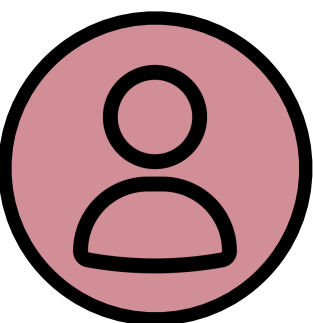
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Implicit Feedback

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CoRLL Algorithm:

coactive learning for large language models

Algorithm: Coactive RLHF

Input: π_1 (initial policy), π_{SFT} (base policy), T (number of rounds), $\mathcal{D}_1 \leftarrow \emptyset$

for $t = 1$ **to** T **do:**

Receive prompt x_t

$y_t \leftarrow \arg \max_y R_{\pi_t}(x_t, y)$

Argmax over MC samples from π_t

Observe improved feedback $\bar{y}_t$

$\mathcal{D}_{t+1} \leftarrow \mathcal{D}_t \cup \{(x_t, y_t, \bar{y}_t, 1)\}$

$\pi_{t+1} \leftarrow \text{DPO}(\mathcal{D}_t, \pi_t, \pi_{\text{SFT}})$

DPO instead of full RLHF pipeline

return π_{T+1}

Experiments

Experiment Setup

Complete IMDB reviews with positive sentiment

Model

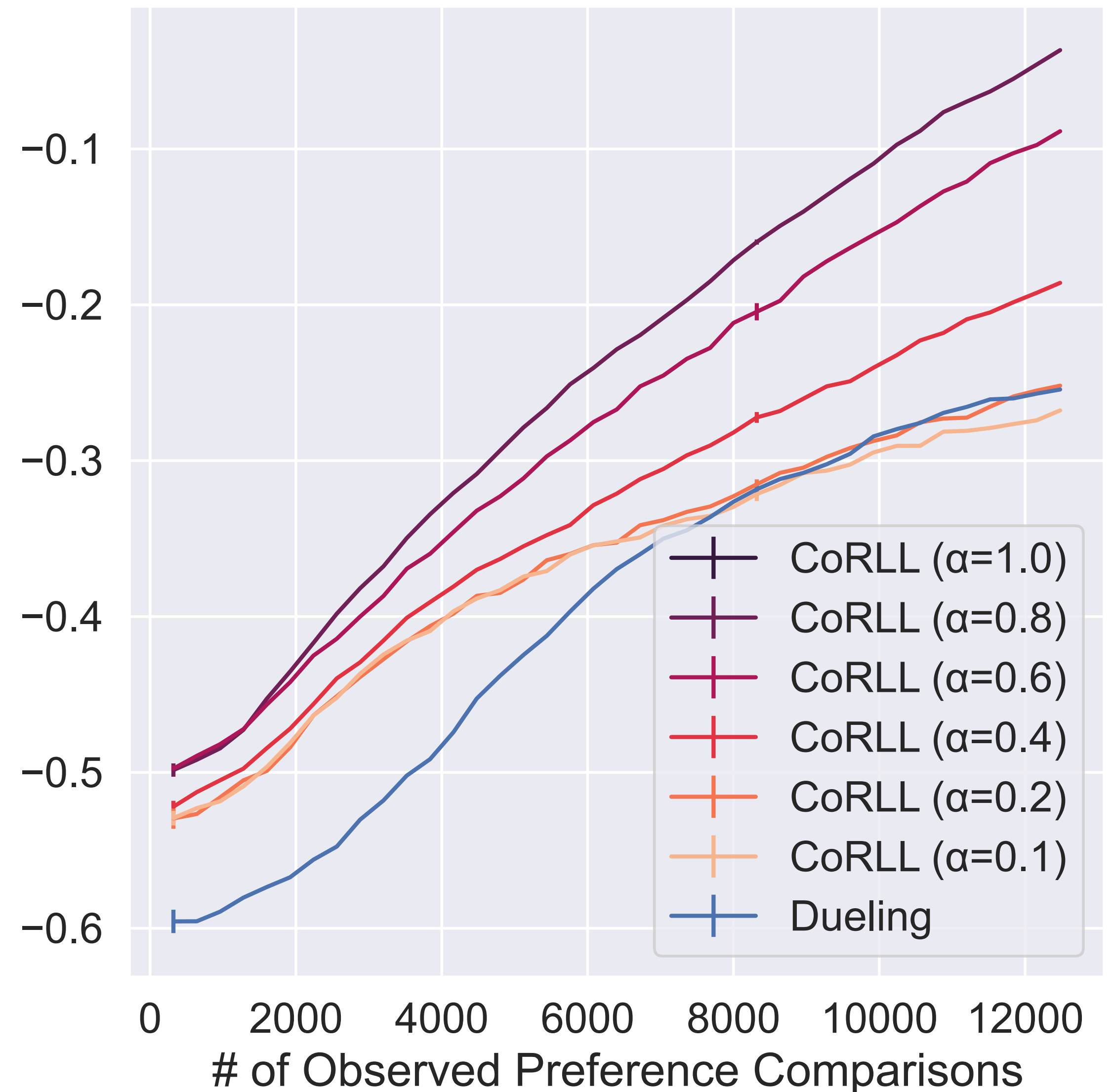
Simulate human preferences with gold-standard LLM

Feedback

α -informative feedback obtained by getting improved $\bar{y}$ from the gold-standard policy

Evaluation

Reward on test examples



Thank You!



Thorsten Joachims



Aaron Tucker



Adam Cahall